

Hamming(7,4): Inspect and Repair

Name: _____

Date: _____

1. Give every position an address

EASY

The three inspector answers are written in the order Inspector 4, Inspector 2, Inspector 1.

Position	Binary address	Inspector 1?	Inspector 2?	Inspector 4?
1	001	yes	no	no
2	010	no	yes	no
3	011	_____	_____	_____
4	100	_____	_____	_____
5	101	_____	_____	_____
6	110	_____	_____	_____
7	111	_____	_____	_____

Why does **000** mean “no error”?

Why are positions 1, 2, and 4 natural places for the check bits?

2. Place data and checks

EASY

Fill the roles in order. Use P for a parity/check bit and D for a data bit.

1	2	3	4	5	6	7
_____	_____	_____	_____	_____	_____	_____

Complete the watcher sets:

Inspector 1 watches _____; Inspector 2 watches _____; Inspector 4 watches _____.

3. Encode report 1010

EASY

Put data **1010** into positions 3, 5, 6, and 7. Choose P_1, P_2, P_4 so every inspector sees an even number of 1s.

P_1	P_2	D_1	P_4	D_2	D_3	D_4
		1		0	1	0

Inspector calculations: $P_1 =$ _____; $P_2 =$ _____; $P_4 =$ _____.

Seven-bit codeword: _____.

4. Locate one damaged bit**EASY**Received word: **1011000**. Test even parity.

Inspector	Positions checked	Satisfied or complains?
1	1, 3, 5, 7	
2	2, 3, 6, 7	
4	4, 5, 6, 7	

Add the labels of complaining inspectors: _____. Error position: _____.

Flip that bit. Repaired codeword: _____. Extract positions 3, 5, 6, 7: _____.

Use the codebook on page 3. Report: _____.

5. Encode and repair independently**CHALLENGE**Encode data **0110** using even parity.

1	2	3	4	5	6	7
_____	_____	_____	_____	_____	_____	_____

Your codeword: _____.

A classmate flips one bit. Received word: _____.

Without being told the position, calculate the syndrome and repair it.

6. Two errors can imitate one**CHALLENGE**Start with valid codeword **1011010**. Flip positions 2 and 5.

- Received word: _____
- Which inspectors complain? _____
- Where does the ordinary Hamming decoder think the error is? _____
- What happens if it “repairs” that position?

- Explain why Hamming(7,4) promises one-error correction, not two-error correction.

7. Add one overall parity bit**CHALLENGE**

An eighth overall parity bit creates SECDED: single-error correction, double-error detection. Complete the decisions.

Seven-bit syndrome	Overall parity	Decision
0	OK	
nonzero	wrong	
0	wrong	
nonzero	OK	

Original NILA Mission Codebook

Decode the four data bits after repairing the seven-bit word.

Code	Mission report	Code	Mission report
0000	Systems normal	1000	Liquid water detected
0001	Battery low	1001	Organic molecules detected
0010	Camera ready	1010	Possible biological pattern
0011	Camera damaged	1011	Mission emergency
0100	Drill ready	1100	Radiation spike
0101	Drill jammed	1101	Surface crack opening
0110	Ice sample collected	1110	Warm plume detected
0111	Salt detected	1111	Return to shelter

Hamming(7,4) reference: check positions are 1, 2, 4; data positions are 3, 5, 6, 7. Inspector 1 watches 1,3,5,7. Inspector 2 watches 2,3,6,7. Inspector 4 watches 4,5,6,7.